For example, you could write a PromQL query that populates a dropdown list for a new `handler` variable, which you could then use in the extended parameterized PromQL query `prometheus_http_requests_total{code="$test",handler="$handler")`. For another example of a query against a data source to populate the values of a dashboard variable, think of a Kubernetes cluster in which you want to provide your dashboard users with a selection of available namespaces they can choose from, based on a PromQL query.

> **TIP** As with data sources, you can provision Grafana with dashboards, either directly, as shown previously, or using Terraform (http://mng.bz/Zq8P). There are many great, community-maintained dashboards available via the Grafana dashboard registry (https://grafana.com/grafana/dashboards/) that you can import by reference or via downloading the JSON definitions.

Alerts (https://grafana.com/docs/grafana/latest/alerting/) are the third core Grafana resource. There have been fundamental changes to the topic of unified alerting, introduced in Grafana version 9. We won't discuss alerts in this chapter but, rather, dive deep into the topic in chapter 7.

You don't have to operate Grafana yourself. If you prefer to outsource the heavy lifting of upgrades, security patches, and access controls, there are many choices (at the time of publication):

- Amazon Managed Grafana (https://aws.amazon.com/grafana/) has been around since late 2020 and went into general availability (GA) in September 2021. It comes with support for AWS SSO and SAMLv2, making it easy to give access to the dashboards to any user in your organization.
- Azure Managed Grafana (http://mng.bz/RxYK) has been in GA since August 2022, supporting Grafana version 9 features.
- You can use managed Grafana as part of Grafana Labs' Grafana Cloud (https://grafana.com/products/cloud/) offering, along with a range of other OSS-based tooling, including Grafana Loki for logs and Grafana Tempo for traces.
- Alibaba Cloud offers Managed Service for Grafana (https://www.alibabacloud.com/product/grafana) that provides integration with collaboration tools, including DingTalk and WeCom.

OK, that was a lot of fun. Next up is Kibana (and its OpenSearch variant, Dashboards).

### 6.1.2 *Kibana and OpenSearch Dashboards*

We discussed Elasticsearch and OpenSearch (ES/OS) as backends in chapter 5. In this context, you have the choice between Kibana (https://www.elastic.co/kibana/), for Elasticsearch and OpenSearch Dashboards (https://opensearch.org/docs/latest/dashboards/index/) for OpenSearch as the respective frontends.